KHIRANA DWICAHYO

Tarakan, North Kalimantan| +62 822 7838 0263 | ranamain28@gmail.com | linkedin

PROFILE

Data Science and Business Intelligence professional with a background in Informatics. Experienced in building machine learning models to identify high-risk borrowers and creating executive KPI dashboards through project-based internships at Bank Muamalat and ID/X Partners. Skilled in Python, SQL (PostgreSQL), and data visualization tools including Power BI, Tableau, and Looker.

EDUCATION

University of Islam Indonesia — **Sleman, Yogyakarta**

Bachelor of Informatics (GPA : 3.58) — *Graduated : November 2023*

- Organization/Awards : APTA UII, Cum Laude, International Journal Publication (Sinta 4 Indexed).

INTERNSHIP EXPERIENCE

Bank Muamalat (Rakamin Academy) | [View Presentation] — **Remote**

BI Analyst (Project Based Internship) — *October 2025 - November 2025*

- Built a master table in PostgreSQL by joining 4 relational tables (Orders, Products, Product Category, Customers), consolidating 2020–2021 transactional data from 2,321 users for analysis.
- Designed an interactive sales dashboard in Looker, tracking revenue, quantity, and category/city-level breakdowns to support executive-level reporting.
- Identified a 4% YoY decline in total sales driven by a 29% drop in Robots category sales in October, and a 65% YoY sales spike in June driven by strong Drones and Robots performance — translating findings into 3 actionable recommendations (seasonal promotions, targeted advertising, product bundling).

ID/X Partners (Rakamin Academy) | [View on Github] — **Remote**

Data Scientist (Project Based Internship) — *January 2025 - February 2025*

- Cleaned, transformed, and encoded raw customer data in Python, preparing a production-ready dataset for credit risk modeling.
- Applied SMOTE oversampling to address class imbalance in the credit dataset, improving minority-class representation for model training.
- Benchmarked 5 classification models (XGBoost, Random Forest, Decision Tree, AdaBoost, Logistic Regression) via 5-fold cross-validation; selected XGBoost as best performer with 99.3% accuracy, precision, and recall (macro-averaged) for early identification of high-risk borrowers.
- Conducted feature importance analysis using XGBoost, identifying "Recoveries" as the most influential variable in the model.

PROJECTS PORTFOLIO

TelcoX Study Case | [View Presentation]

Self-initiated case study simulating a telecom analytics engagement, using AI-generated data to model customer usage behavior and ARPU optimization.

- Extracted and integrated data from 5 relational tables (Customer Profile, Usage Activity, Billing, Subscription Plan, Add-on Usage) via SQL to analyze usage behavior across 2,000 customers.
- Identified a 38% data overusage rate and a consistent mismatch between customer usage and subscribed plan quotas across all usage segments, revealing that ARPU was driven more by plan tier (Rp60K–Rp188K) than by actual usage intensity.
- Delivered 5 data-driven recommendations (targeted upsell, plan bundling, usage-based nudges) to realign customers with suitable plans, projecting incremental ARPU growth from upgrading misaligned heavy users.

Worldometers 2025 Analysis | [View on Github] · [View on Looker]

Project analyzing global population growth trends and demographic shifts using web-scraped data.

- Built a two-stage Scrapy web crawler to extract historical demographic data (1955–2025) across 200+ countries from Worldometers, capturing 13 metrics per country including population, fertility rate, migration, and urbanization.
- Analyzed global population trends in Looker, identifying India's overtake of China as the world's most populous country in 2023 and China's shift into negative population growth (-0.23%) since then.

- Uncovered a negative correlation between urbanization and population growth rate, and linked declining global fertility (5.41 → 2.31 children per woman, 1955–2025) to slowing growth, with Sub-Saharan Africa remaining the primary driver of high fertility and growth.

Pima Indians Diabetes Prediction | [View on Github]

A machine learning project predicting diabetes risk using clinical diagnostic measurements.

- Cleaned clinical data with domain-informed missing value handling (KNN imputation for Glucose/BMI/BloodPressure, median imputation for Insulin) and outlier capping based on IQR analysis.
- Engineered clinically meaningful features by converting raw measurements into diagnostic categories (e.g., BMI into Underweight/Ideal/Overweight/Obese, Glucose into Normal/Prediabetes/Diabetes) to support interpretability.
- Benchmarked 4 classification models (Random Forest, XGBoost, AdaBoost, Decision Tree) evaluated via accuracy, precision, recall, and F1-score; selected Random Forest as best performer (75.1% accuracy, 75.1% recall) using ROC and precision-recall curve analysis.

SKILLS & INTEREST

Skills : Programming (Python, SQL/PostgreSQL, R) | Machine Learning (scikit-learn, XGBoost, imbalanced-learn) | Data Visualization (Power BI, Tableau, Looker) | Microsoft Excel
Interest : Business Intelligence & Dashboarding, Predictive Analytics & Machine Learning, Data Warehousing, Web Scraping & Data Engineering

ADDITIONAL INFORMATION

Languages : Indonesia (native), English, Korean.
Certifications : Web Scraping (Python, Selenium, Scrapy) – Udemy; Mastery Deep Learning – Digital Talent Scholarship.